

UM12505

FRDM-A-S32K312 development board for MCUs hardware user manual

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User manual

Document information

Information	Content
Keywords	FRDM-A-S32K312, S32K3, Automotive, cost effective board, automotive, general purpose MCU, Automotive MCU, S32K312, Automotive MCU, 32 bit automotive MCU, ARDUINO compatible, CAN, FS26, ARDUINO shield
Abstract	Development board for S32K312 MCUs hardware user manual



1 Introduction

The NXP FRDM-A-S32K312 is designed to explore the capabilities of the NXP S32K312 automotive-grade Microcontroller Unit (MCU). The FRDM-A-S32K312 is based on the full-featured S32K312EVB-Q172, which is an ideal tool for engineers working in automotive, embedded, and industrial systems, as well as those seeking for a compact and versatile development solution.

This board features a USB Type-C port and a Power JACK connector to enable two ways to supply the board depending on the application and requirements. Even being supplied via USB-C, the boards support communications protocols that require a +12 V supply due to the integrated NXP FS26 system basis chip (SBC). This SBC includes a boost converter that steps up the input voltage to meet the requirements of these protocols.

The FRDM-A-S32K312 can be used for prototyping, architectural learning, and custom application development.

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2 Finding kit resources and information on the NXP web site

NXP Semiconductors provides online resources for this evaluation board and its supported device(s) on <https://www.nxp.com/>.

The information page for the FRDM-A-S32K312 evaluation board is at <https://www.nxp.com/FRDM-A-S32K312>. The information page provides overview information, documentation, software and tools, parametrics, ordering information and a **Getting Started** tab. The Getting Started tab provides quick-reference information applicable to using the FRDM-A-S32K312 evaluation board, including the downloadable assets referenced in this document.

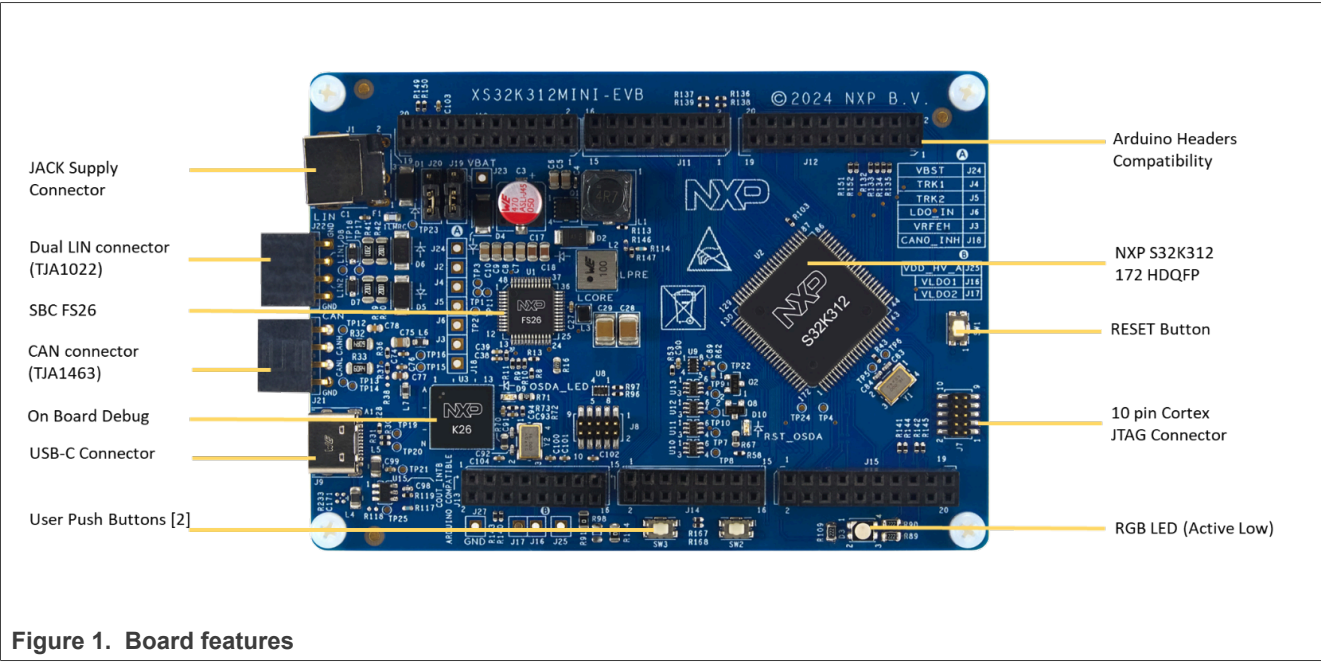
3 Getting ready

3.1 Kit contents

- Board
- USB Type-C cable
- Getting started postcard
- Kit content sheet

4 Getting to know the hardware

4.1 Board features



4.2 Block diagram

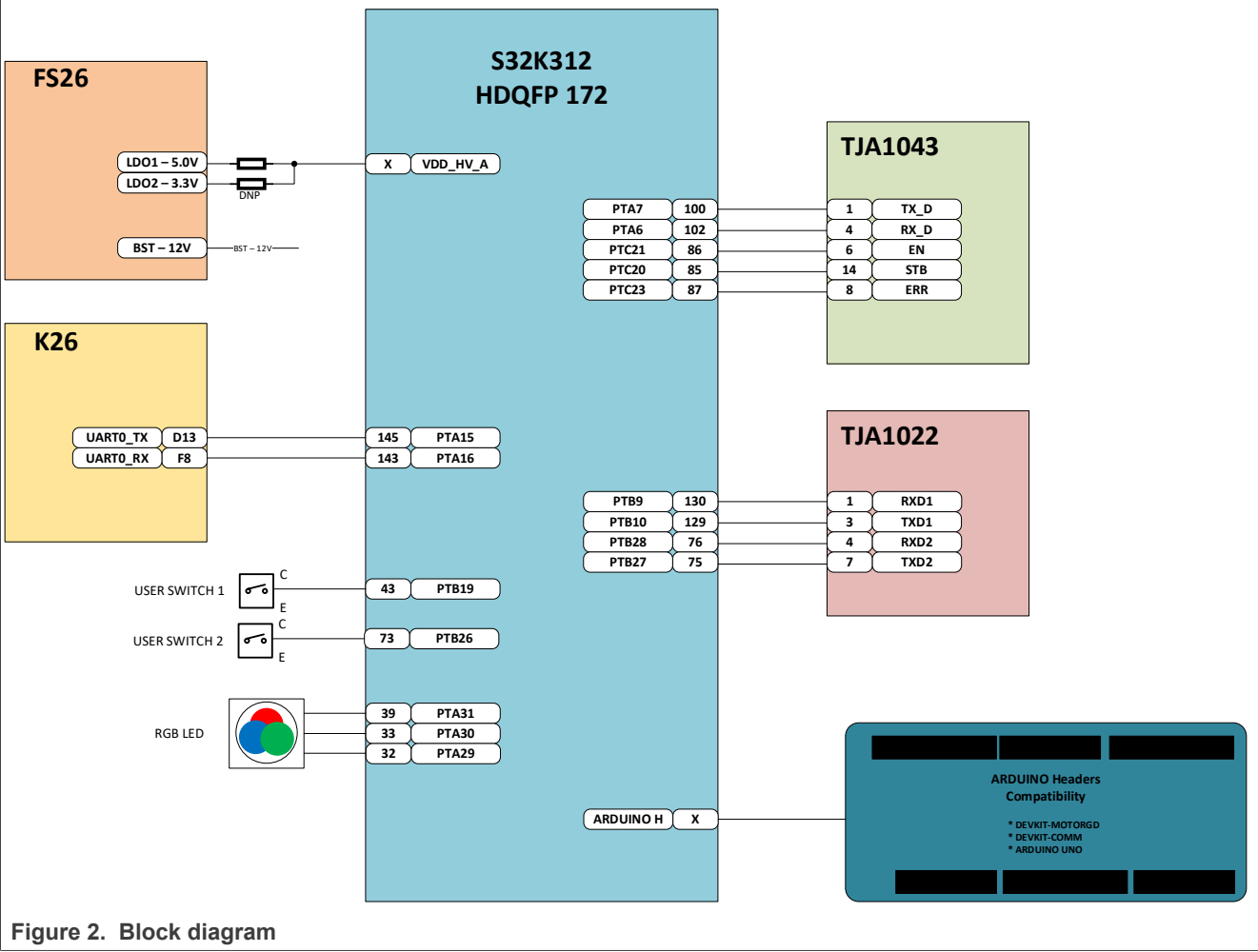


Figure 2. Block diagram

4.3 Kit featured components

4.3.1 Power supply

4.3.1.1 Supply options

The FRDM-A-S32K312 is a fast prototyping board based on the S32K312EVB-Q172, which means that this product was not designed as an end board. The board includes two ways to be supplied according to the application, which are described in [Table 1](#).

Table 1. Supply selector

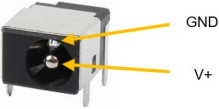
Jumper position for J19 and J20	Mode	Max voltage/ current supported	Description
	Low demand current application	+5 V/400 mA	The first method to provide power is via the J9 USB-C connector ^[1] , but the current is limited to 500 mA ^[2] . This supplying method is only recommended for low demand current applications. Even in this supplying method, CAN or LIN PHY can work because the FS26 enables a boost circuit to supply the +12.0 V voltage references of the PHYs. ^[3] The maximum current demand for +12.0 V in this supplying method is 170 mA due to the boost circuit consumption. Follow the steps in the Table 4 to supply the board with the above described methodology.
	High demand current application	+12 V/2 A	The second method to supply power requires plugging in the voltage supply in the JACK connector J1 (+12.0 V/2 A), this option is recommended for applications with higher demand current or when using compatible boards. Follow the steps in the Table 4 to supply the board with the above described methodology.

[1] This board does not include power delivery connected to a USB-C connector input.

[2] This is to enable for desktops with USB-A or USB-C connectors.

[3] It is not recommended to use both PHYs at maximum current demand at the same time.

Table 2. Supply connectors

Supply Connector	Reference	Description
	J1	2.1 mm DI 5.5 mm DE barrel connector This connector should be used to connect the supplied wall-plug mains adapter. Note: If a replacement or alternative adapter is used, make sure the 2.1 mm plug uses the correct polarization as shown.
	J9	CONN RCP USB2.0 TYP C 24P SMD USB 2.0 TYPE-C receptacle connector 24 (16 + 8 Dummy) position surface mount, right angle; through hole

4.3.1.2 Startup sequence

Table 3. Supply board method (low current)

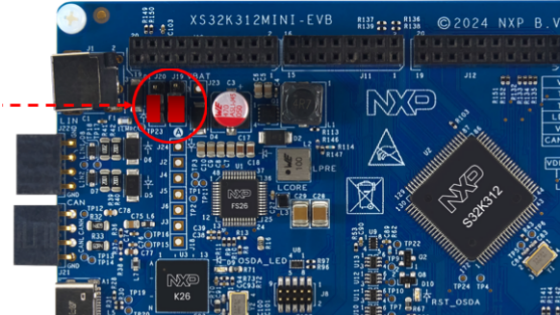
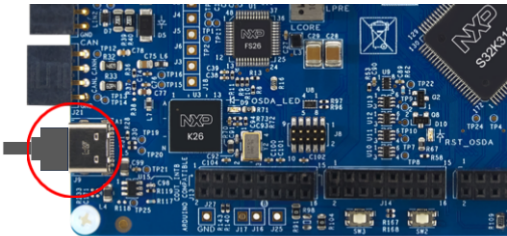
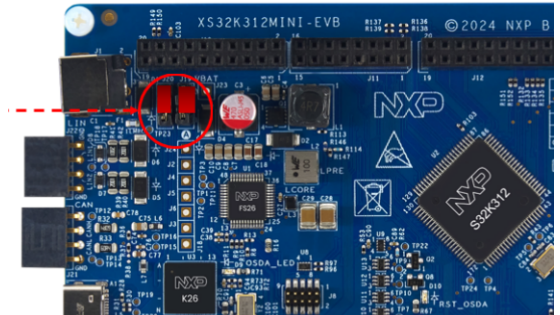
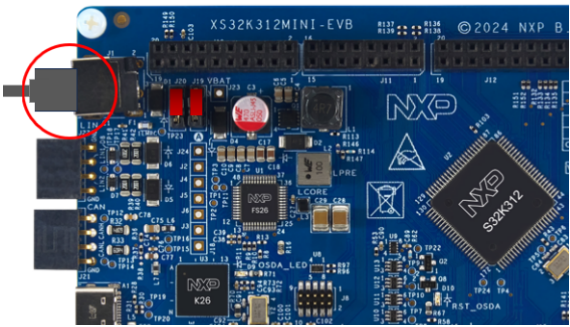
Step sequence	Description	Graphical description
A)	Ensure that the jumpers J19 and J20 are placed in the position 1 – 2 to power supply via the USB-C connector (J9).	
B)	Plug in the USB-C cable to the J9 connector	

Table 4. Supply board method (high current)

Step sequence	Description	Graphical description
A)	Ensure that the jumpers J19 and J20 are placed in the position 2 – 3 to power supply via JACK connector (J1).	
B)	Plug in the 12 V power supply through J1 connector	

4.3.1.3 Voltage references

The FRDM-A-S32K312 has two possible options to supply power to the board, the FS26 manage the input voltage to supply the different voltage domains in the layout. The PMIC can have different voltage values for the two different input voltage values.

When the board is supplied through the USB-C connector, the FS26 uses the internal boost to increase the voltage from +5.0 V to +12.0 V. This voltage ranges are required for the LIN and CAN PHYs to have a correct functionality even when using 5 V as main power supply (via J9). In this mode, the maximum power consumption the board can manage is 150 mA at the output of the boost.

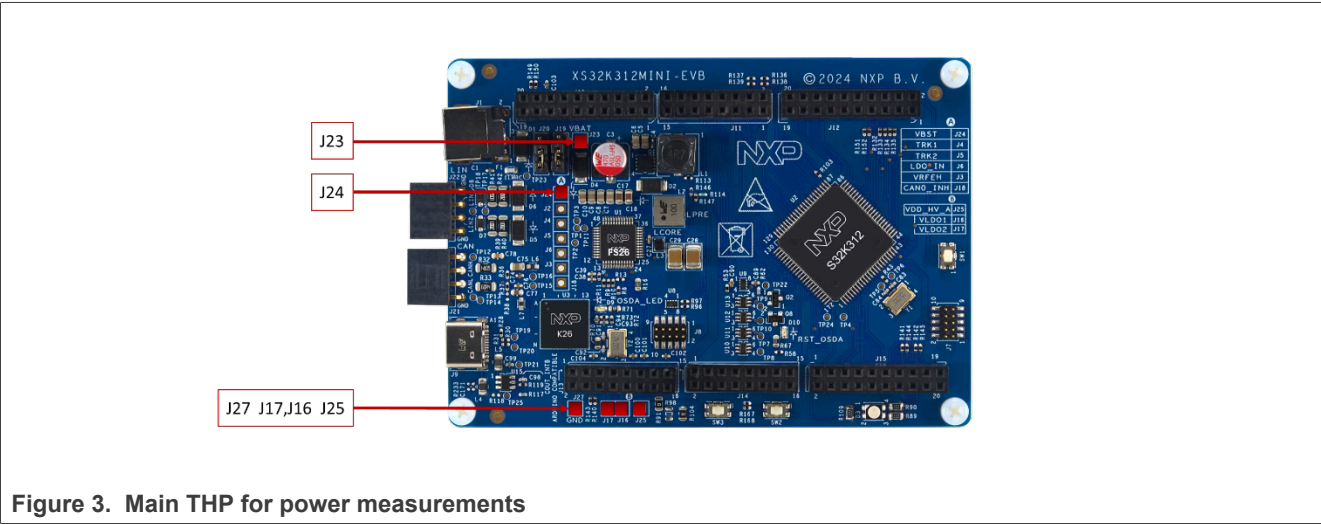
When the boost is not necessary because the board is supplied via J1 connector for more current demand applications, the FS26 boost does a bypass of the voltage from the power source. In this case, the boost is automatically disabled and the +12.0 V voltage references of the board are supplied from the JACK connector.

The FS26 integrates two LDOs. These LDOs are programmed to generate +5.0 V on LDO1 and +3.3 V on LDO2 output. This output voltages supply the MCU and the peripherals connected to it.

For voltage measurement, this board includes some through hole pads to measure the main voltage references as VBAT, VBST, P5V0 (FS26 LDO1), P3V3 (FS26 LDO2), VDD_HV_A, and GND (see [Table 5](#)).

Table 5. Through hole pads for measure voltages

Image	THP reference	Signal	Expected voltage	Description
1 2 3	J16	P5V0	+5.0 V	Output voltage from LDO1 of the FS26, the current is limited to 400 mA and supplies
	J17	P3V3	+3.3 V	Output voltage from LDO2 of the FS26, the current is limited to 400 mA
	J23	VBAT	+5.0 V	In case the low-current option are selected by jumpers J19 and J20
			+12.0 V	In case the high-current option are selected by jumpers J19 and J20
	J24	VBST	+12.0 V	Output voltage from FS26 Boost. This reference is not used when the high-current option is selected from supply jumpers.
	J25	VDD_HV_A	+5.0 V/+3.3 V	This is the voltage domain that supplies the S32K312 MCU. This voltage can vary according to the resistor populated, but some peripherals or interfaces as RGB LED and CAN do not support the +3.3 V supply.
	J27	GND	–	Ground reference of the board



4.3.1.3.1 Main supply

The FRDM-A-S32K312 includes two ways to be supplied to enable both high and low demand current.

When a low demand current is selected using the required configuration in jumpers J19 and J20, the voltage is supplied from the USB-C connector J9 and passes through a current limiting circuit (MIC2005-0.5YM6) to avoid demanding more current to the connected desktop USB port.

Then, using the J20 in position 1 – 2, the voltage reference is named VBAT, which is connected to the FS26 boost circuitry. This increases the voltage from +5.0 V to +12.0 V. With this value, the voltage reference is named VBST, but works in any boost circuit. The current consumption increases to generate the required voltage.

In case the high current supply option is selected by the user in J20 2 – 3 position, the current flows from J1 through the fuse F1 for the signal reference to become in VBAT. In this case, the FS26 identifies that the boost is not required, so the 12 V only supplies the VPRE signal in the FS26. Refer to the schematics for more details.

4.3.1.3.2 FS26 supply

All the voltage and current of this board is managed by the FS26, which is an automotive safety system basis chip (SBC), designed to support entry and mid-range safety microcontrollers, like S32K312.

FS26 devices have multiple power supplies and the flexibility to work with other microcontrollers targeting automotive electrification. Possible FS26 applications include power train, chassis, safety and low-end gateway technology.

The FS26 features multiple switch mode regulators and low dropout (LDO) voltage regulators to supply the microcontroller, sensors, peripheral ICs, and communication interfaces. Due to this, SBC integrates two LDO outputs the MCU can be supplied by 3.3 V or 5.0 V (These voltages are programmed by the OPT of the SBC.). The maximum current that can be provided by these LDOs are 400 mA, however, the connected interfaces do not require more than that current. Refer to the schematics for more details.

4.3.1.3.3 VDD_HV_A

This is the main I/O and analog supply voltage for the S32K312 MCU, in this case the MCU is supplied by 5.0 V output voltage from the FS26 LDO1. A decoupling capacitor is required for each voltage pin and all the capacitors need to be routed to the same voltage reference with the only consideration that each capacitor must be placed near of its voltage pin. The capacitance of decoupling capacitors is included in the S32K3 hardware design guidelines.

Additionally, a bulk capacitor is recommended to increase the MCU supply voltage robustness. Refer to the schematics for more details.

The VDD_HV_A voltage is used internally to generate the required voltage for the core and other circuitry. This internal circuitry is included to generate the voltage reference for the output voltage for GPIO pins (+5.0 V as default) and other features that include this MCU.

In this board, the pin VREFH of the MCU is connected to VDD_HV_A due to that pin is the ADC high reference voltage, and the maximum voltage at the input of a pin is 5.0 V.

4.3.1.3.4 V11

Although this MCU does not require an external power supply in the pins of V11, it is mandatory to connect all the pins of V11 together, but in each of them connect a decoupling capacitor to keep integrity in the core voltage that is generated internally from the VDD_HV_A voltage reference. Refer to the schematics for more details.

4.3.1.3.5 V25

The V25 pin is another pin, which voltage is generated internally by the VDD_HV_A. This supplies the flash memory and (via double bond) the clock modules, however, a decoupling capacitor near of the pin is mandatory for this voltage reference. Refer to the schematics for more details.

Table 6. Decoupling and bulk capacitors used in FRDM-A-S32K312

Capacitor	Characteristic	Value
Decoupling capacitor	X7R/X8R ceramic	100 nf
Bulk capacitor	X7R/X8R ceramic	10 uF

4.3.2 Programming and debug interface

4.3.2.1 RESET switch and LED indicator

The RESET switch [SW1] provides for manual application of the RESET input signal. The S32K3 MCU drives the RESET signal to reset the board peripherals. The RESET LED indicator [D10] is on during the reset signal. This operation indicates the S32K312 MCU is in the Reset state.

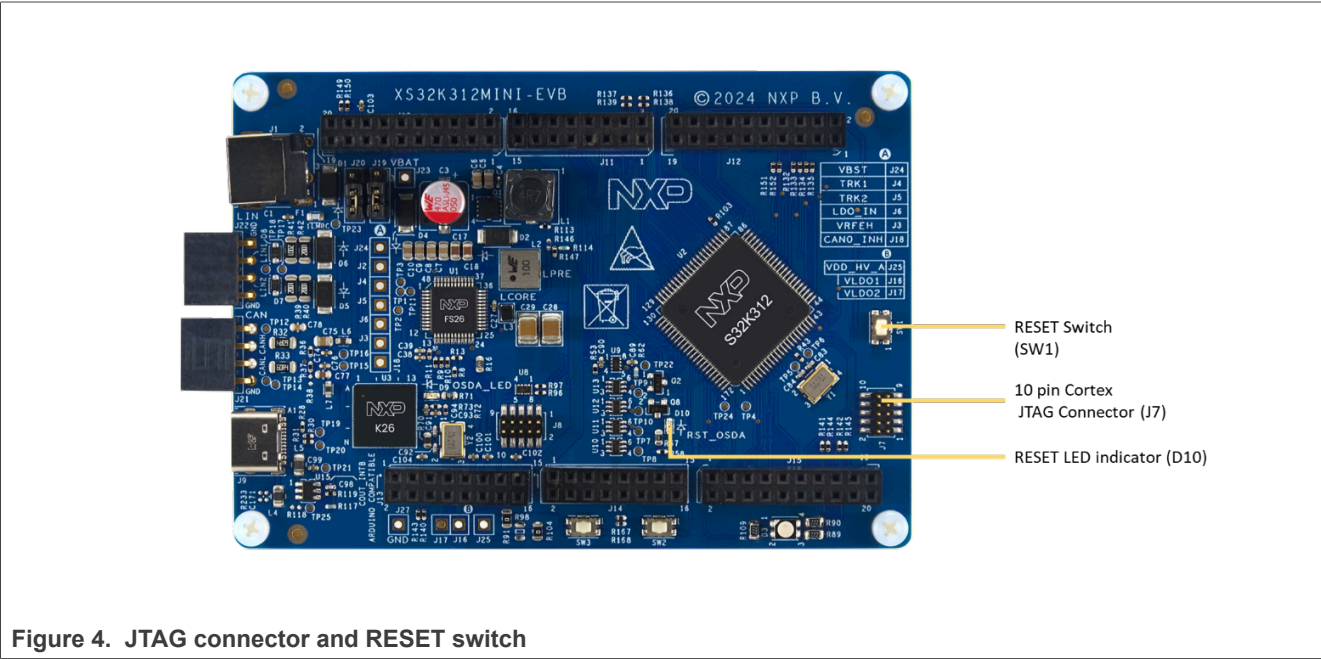
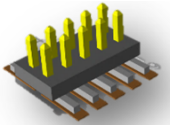


Figure 4. JTAG connector and RESET switch

4.3.2.2 JTAG

The FRDM-A-S32K312 incorporates JTAG connectors to program the MCU using a PE Multilink Universal FX Debug Probe, which allows the user control the target’s execution, read/write registers and memory values, debug code on the processor, and program internal or external flash memory devices. In summary, it bridges serial and debug communications between a USB host and an embedded target processor. Refer to the schematics for more details.

Table 7. Programming and debug connectors

Connector		Reference/ component	Description
Ten-pin cortex debug		J7	The cortex debug connector provides support for serial wire and JTAG interface modes in a very small, and low-cost ten-pin (0.05 ") connector. This new connector style provides access to all SWD, SWV, and JTAG signals available on a Cortex-M device. A ten-pin header (Samtec FTSH-105-01) is specified with these dimensions: 0.25 " x 0.188 " (6.35 mm x 4.78 mm).

4.3.2.3 On-board debugger

The board incorporates an on-board debugger embedded as JTAG connectors. It bridges serial and debug communications between a USB host and an embedded target processor (that is, the K26).

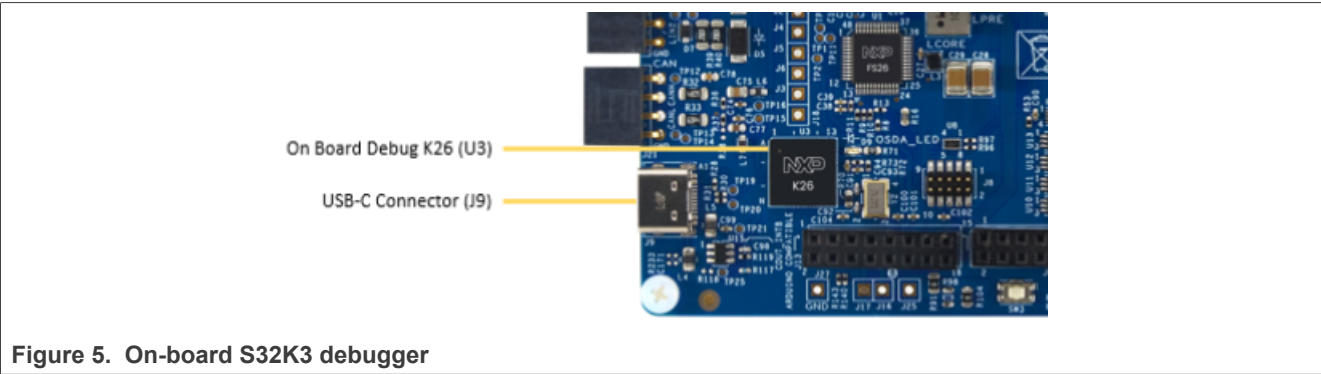


Figure 5. On-board S32K3 debugger

4.3.3 LIN interface

The FRDM-A-S32K312 incorporates two LIN interfaces connected to the S32K312 MCU. Using NXP LIN transceivers from TJA1022, supporting controller mode. The output from the LIN transceivers is connected to J22. The pinout of these headers is shown in [Table 8](#). Refer to the schematics for additional information.

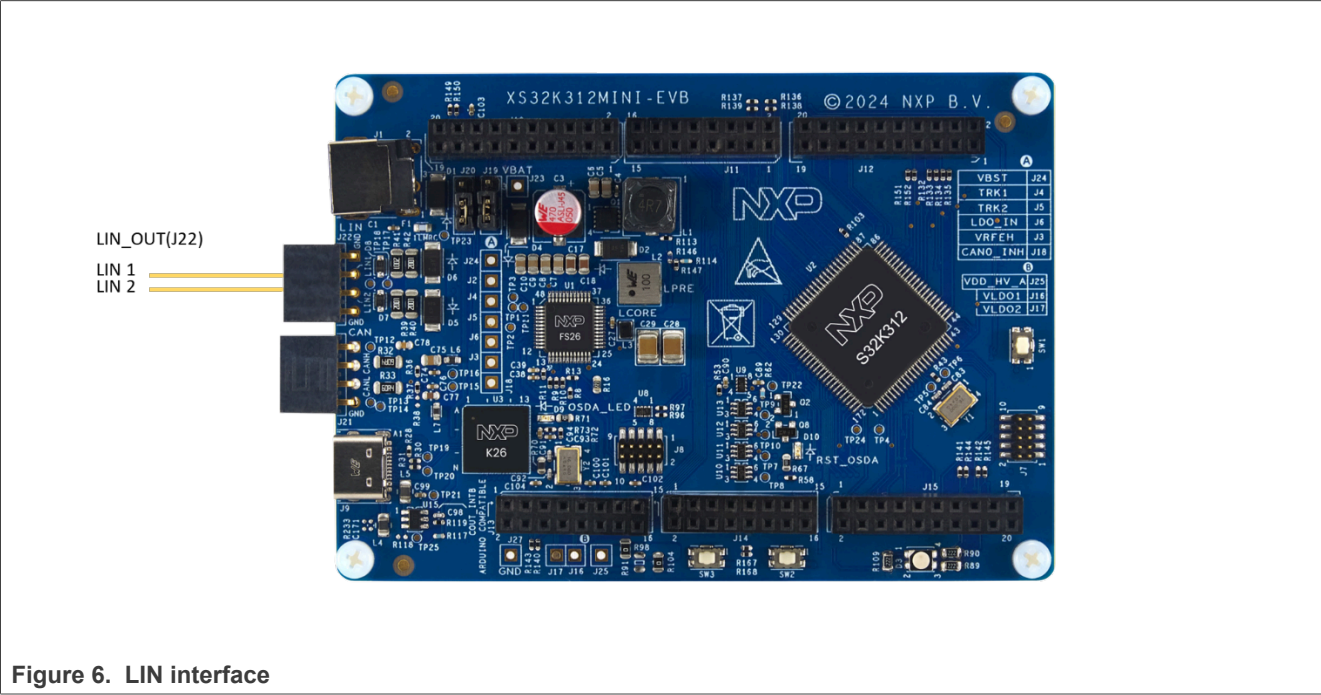


Table 8. LIN output connection

Connector	Reference	Pin number	Signal
	J22	1	GND
		2	LIN1
		3	LIN2
		4	GND

One of the greatest features of this board is the capability to use LIN interface with or without 12 V supply using the boost from the FS26. In this case, the 12 V supplies the LIN PHY TJA1022 pin 10 (VBAT) and the LIN out signals. Do not use the LIN and CAN PHY while the board is supplied by the low-current methodology (that is, using USB connector J9).

Table 9. LIN signal assignment

LIN interface	Signal name	MCU port	Comments
TJA1022	LIN1_RX	PTB9	FXIO D28
	LIN1_TX	PTB10	FXIO D27
	LIN2_RX	PTB28	LPUART5_RX
	LIN2_TX	PTB27	LPUART5_TX

4.3.4 CAN interface

This board uses an NXP CAN transceiver; TJA1043. The output from the CAN transceiver is connected to J21. See details in the [Table 10](#).

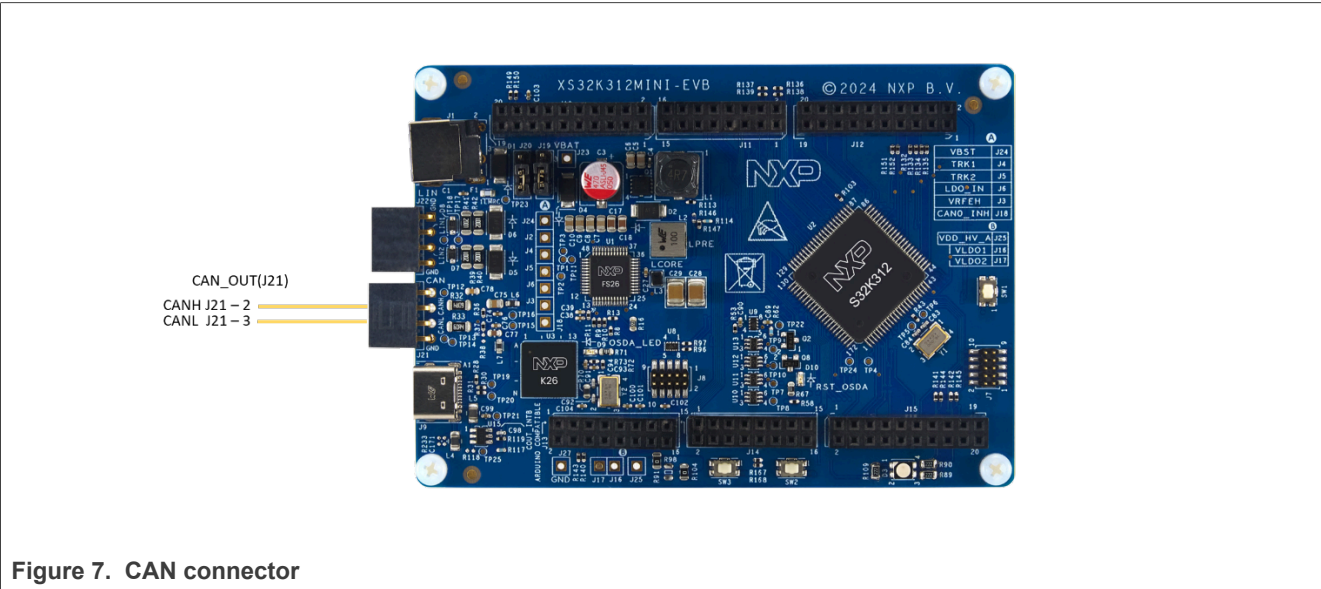


Table 10. CAN interface – Connector

Connector	Reference	Circuit/ interface	Pin number	Signal/connection
	J21	CAN0	1	NC
			2	CANH
			3	CANL
			4	GND

This CAN PHY can be supplied by the high-current demand method and the low-current demand method, but only use this PHY in case the low-current demand method is under use. This differential pair is routed at 100 Ω. Refer to the schematics for additional information.

The S32K312 MCU has a great feature that includes the FLEXCAN connected to the PHY to improve the communication from the board to another node.

Table 11. CAN interface – MCU connections

LIN interface	Signal name	MCU port	Comments
TJA1043	CAN0_RX	PTA6	[CAN0_RX Module] is routed to CAN Phy0
	CAN0_TX	PTA7	[CAN0_TX Module] is routed to CAN Phy0
	CAN0_ERRN	PTC23	PTC23 is routed to CAN Phy0 as CAN0_ERRN
	CAN0_EN	PTC21	PTC21 is routed to CAN Phy0 as CAN0_EN
	CAN0_STB	PTC20	PTC20 is routed to CAN Phy0 as CAN0_STB

4.3.5 User peripherals

4.3.5.1 RGB LED indicator

The board includes active low-user RGB LEDs connected to the MCU ports. The user LEDs are connected as follows:

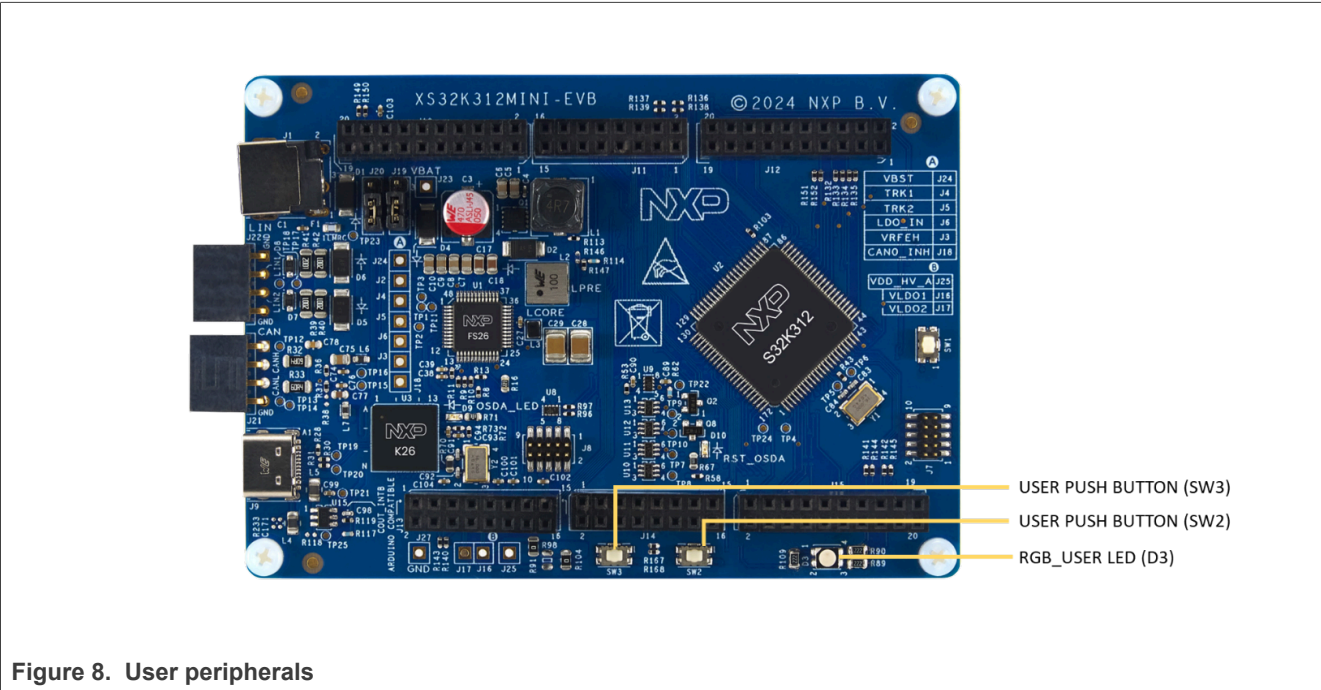
Table 12. USER LED Indicator

Reference	Signal name	MCU port	Comments
D3	RGB_REDLED_PTA29	PTA29	RED LED active low
	RGB_GREENLED_PTA30	PTA30	GREEN LED active low
	RGB_BLUELED_PTA31	PTA31	BLUE LED active low

Warning: If the board is supplied by 3.3 V reference, the LED does not work as expected.

4.3.5.2 User Push buttons

The board integrates two **Push** buttons active low (pulled high), driven to VDD_HV_A, the **Push** button SW2 is connected to PTB19 MCU port while the **Push** button SW3 is connected to PTB26. The switch is connected as follows:



This user **Push** buttons are connected to PTB19 and PTB26 pin in the S32K312MCU, and can be used for debugging or application uses.

Table 13. User Push buttons

Peripheral	MCU port	Comments
SW2	PTB19	Active high Push buttons
SW3	PTB26	

4.3.6 Pinout assignment

Despite the small size of this board and the ARDUINO pin headers reduction present in the FRDM-A-S32K312 compared to S32K312EVB-Q172, this board includes compatibility in the inner connectors with the S32K312EVB-Q172 board. If you build a project in the FRDM-A-S32K312, it can be escalated to the EVB with almost any modification in the pins (Unless one of the pins whose resistance is in DNP is used).

The FRDM-A-S32K312 was designed to be compatible with other NXP boards such as [DEVKIT-MOTORGD](#), which is a plug-in board/shield for motor control solution and [DEVKIT-COMM](#), which is a hardware plug-in adapter board to supports four additional CAN and six additional LIN/SCI connections, to plug in the board to expand its functionalities.

Note: Those boards are not included in the FRDM-A-S32K312.

See [Figure 9](#) for more details related to DEVKIT-MOTORGD and DEVKIT-COMM pinout in this board.

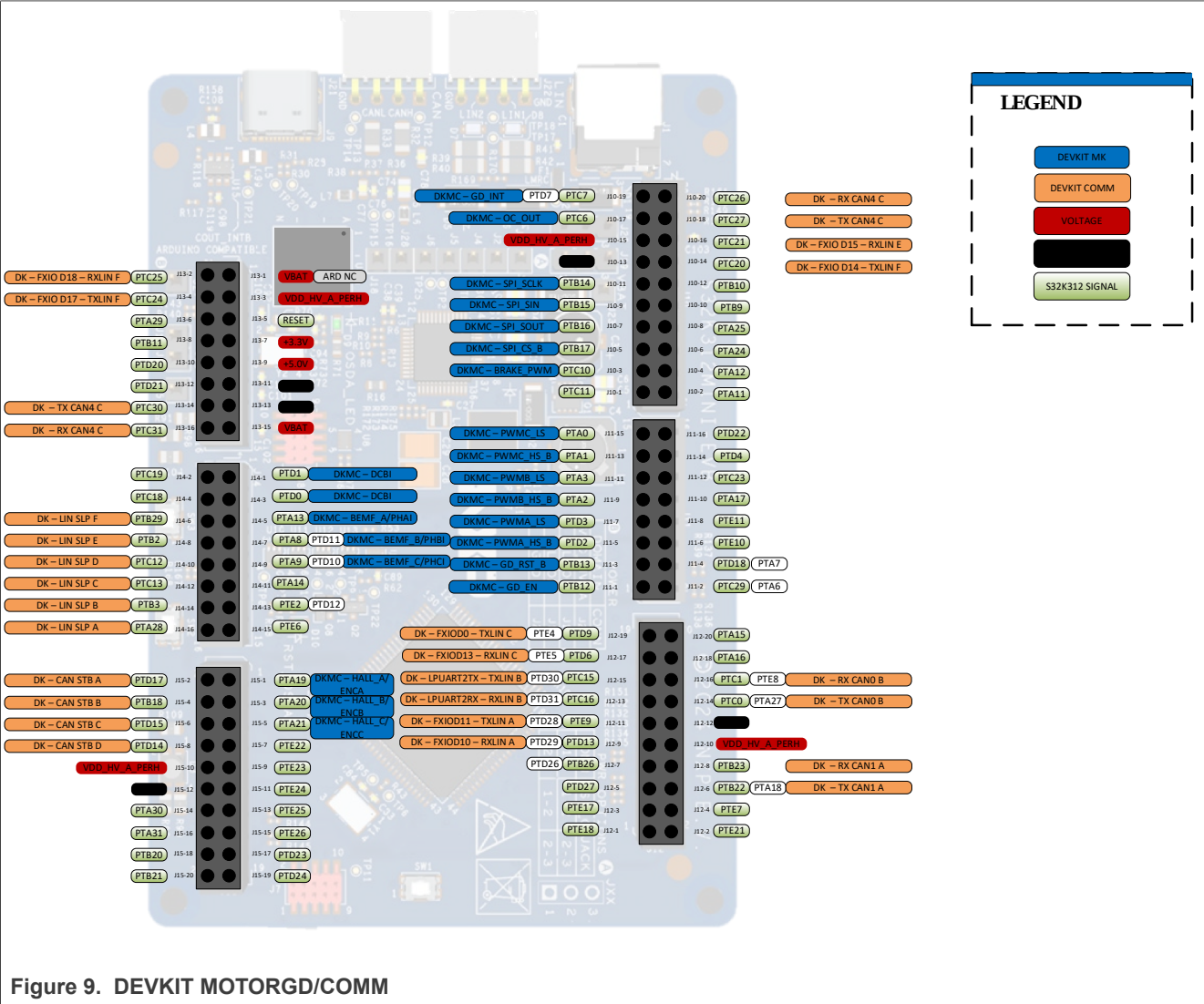
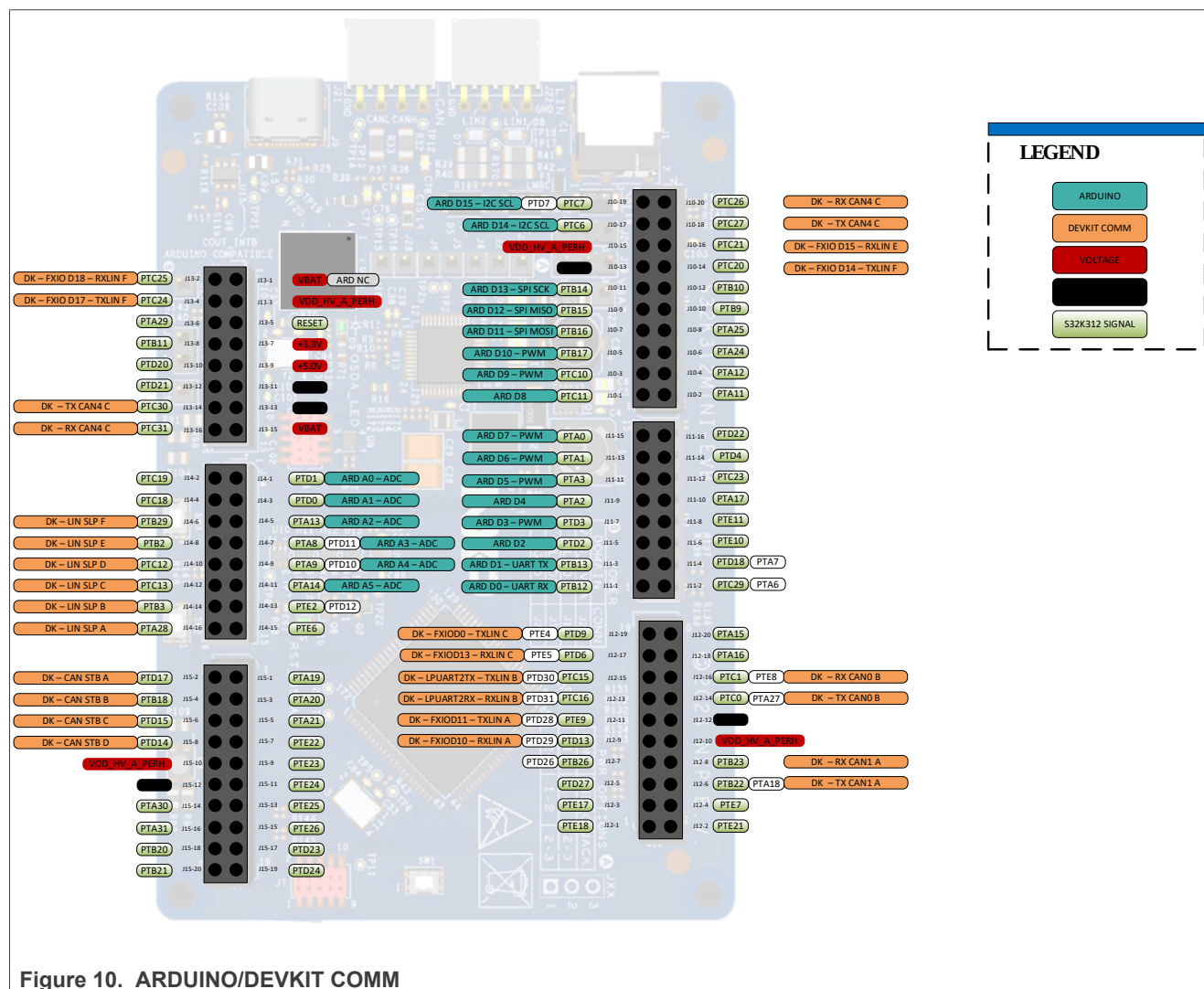


Figure 9. DEVKIT MOTORGD/COMM

Additionally, the FRDM-A-S32K312 includes the ARDUINO UNO standard, see [Figure 10](#).



4.4 Schematic, board layout and bill of materials

The schematic, board layout and bill of materials for the FRDM-A-S32K312 evaluation board are available at <https://www.nxp.com/FRDM-A-S32K312>.

5 Configuring the hardware

5.1 Default configuration

Table 14. Default configuration

Interface	Reference/signal	Default configuration	Description/comment
MCU	U2	S32K312 MCU	ASIL B single ARM M7 core 120 MHz and 2 MB general purpose MCU. Optimized for real-time with zero wait I/D-TCM.
Supply jumpers	J20	1 – 2	Select the supply input. See Section 4.3.1.1 for more details.
	J19	1 – 2	
MCU power supply	VDD_HV_A	+5.0 V/+3.3 V	The actual voltage connected to this reference signal is +5.0 V whose is generated by the FS26 LDO1. It is the main I/O and analog supply voltage for the S32K312 MCU. The +3.3 V is disconnected as default. This voltage is generated by FS26 LDO2.
	V11	+1.1 V	Core logic supply internally generated
JTAG	J7	Enable	The JTAG interface of the S32K312 MCU is connected through a 10 pin 1.27 mm header.
Dual LIN interface	J22	J22-2 – LIN1	Configured by FXIO D28 and D29
		J22-3 – LIN2	Configured by LPUART5
CAN interface	J21	J21-2 – CANH	The S32K312 is connected to a TJA1463 CAN PHY through CAN0 RX and TX signals.
		J21-3 – CANL	
On-board debug	PTA15 – LPUART6_RX	Enable	The on-board debug interface is connected through JTAG pins and LPUART6 interface
	PTA16 – LPUART6_TX	Enable	
User RGB LED	D3	PTA29	Red [Active Low]
		PTA30	Green [Active Low]
		PTA31	Blue [Active Low]
User Push button	SW2	PTB19	For general purpose
	SW3	PTB26	For general purpose
ARDUINO headers	–	–	This board incorporates compatibility with ARDUINO UNO, DEVKIT-MOTORGD and DEVKIT-COMM boards. Refer to the schematics and the ARDUINO chapter for more details in the routed pins.

6 Abbreviations

[Table 15](#) defines the abbreviations used in this document.

Table 15. Abbreviations

Abbreviation	Description
BST	Boost
CCM	Counter with CBC MAC (Cipher block chaining message authentication code)
CMOS	Complementary metal oxide semiconductor
CP	Charge pump
CPU	Central processing unit
CSPI	Configurable serial peripheral interface
DDR	Double data rate
DIP	Dual in-line package
DPGA	Differential programmable gain amplifier
EEPROM	Electrically erasable programmable read only memory
EPROM	Erasable programmable read only memory
FET	Field-effect transistor
GCTL	Gate control
GDU	Gate driver unit
GPIO	General purpose input/output.
GPO	General purpose output.
HG	High-side gate
HS	High-side source
HW	Hardware
HVI	High voltage input
HVM	High voltage module
I ² C	Inter-integrated circuit
I/O	Input/output
JTAG	Joint test access group
LED	Light emitting diode
LG	Low-side gate
LPM	Low-Power mode
LS	Low-side source
MB	Megabyte
MCU	Microcontroller unit
MOSFET	Metal-oxide-semiconductor field-effect transistor
MS	Memory stick
NVRAM	Non-volatile random-access memory

Table 15. Abbreviations...continued

Abbreviation	Description
PCB	Printed circuit board
PHY	Physical interface
PMC	Power management controller
POR	Power-on reset
PSRAM	Pseudo random access memory
PWR	Power
PWM	Pulse width modulation
RAM	Random access memory
SDRAM	Synchronous dynamic random-access memory
TFT	Thin film transistor
UART	Universal asynchronous receiver/transmitter
USB	Universal serial bus

7 References

FRDM-A-S32K312 — detailed information on this board, including documentation, downloads, and software and tools <https://www.nxp.com/FRDM-A-S32K312>

8 Revision history

Table 16. Revision history

Document ID	Release date	Description
UM12505 v.1.0	04 March 2026	Initial public release

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Tables

Tab. 1.	Supply selector	5	Tab. 9.	LIN signal assignment	11
Tab. 2.	Supply connectors	5	Tab. 10.	CAN interface – Connector	12
Tab. 3.	Supply board method (low current)	6	Tab. 11.	CAN interface – MCU connections	12
Tab. 4.	Supply board method (high current)	6	Tab. 12.	USER LED Indicator	13
Tab. 5.	Through hole pads for measure voltages	7	Tab. 13.	User Push buttons	13
Tab. 6.	Decoupling and bulk capacitors used in FRDM-A-S32K312	9	Tab. 14.	Default configuration	16
Tab. 7.	Programming and debug connectors	10	Tab. 15.	Abbreviations	17
Tab. 8.	LIN output connection	11	Tab. 16.	Revision history	18

Figures

Fig. 1.	Board features	3	Fig. 6.	LIN interface	11
Fig. 2.	Block diagram	4	Fig. 7.	CAN connector	12
Fig. 3.	Main THP for power measurements	7	Fig. 8.	User peripherals	13
Fig. 4.	JTAG connector and RESET switch	9	Fig. 9.	DEVKIT MOTORGD/COMM	14
Fig. 5.	On-board S32K3 debugger	10	Fig. 10.	ARDUINO/DEVKIT COMM	15

Contents

1	Introduction	2
2	Finding kit resources and information on the NXP web site	2
3	Getting ready	3
3.1	Kit contents	3
4	Getting to know the hardware	3
4.1	Board features	3
4.2	Block diagram	4
4.3	Kit featured components	5
4.3.1	Power supply	5
4.3.1.1	Supply options	5
4.3.1.2	Startup sequence	6
4.3.1.3	Voltage references	7
4.3.2	Programming and debug interface	9
4.3.2.1	RESET switch and LED indicator	9
4.3.2.2	JTAG	10
4.3.2.3	On-board debugger	10
4.3.3	LIN interface	11
4.3.4	CAN interface	12
4.3.5	User peripherals	13
4.3.5.1	RGB LED indicator	13
4.3.5.2	User Push buttons	13
4.3.6	Pinout assignment	14
4.4	Schematic, board layout and bill of materials	15
5	Configuring the hardware	16
5.1	Default configuration	16
6	Abbreviations	17
7	References	18
8	Revision history	18
	Legal information	19

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